Fiscal Year:	FY 2021	**Task Last Updated:**	FY 09/13/2021
PI Name:	Cromer, Walter Ph.D.		
Project Title:	Use of Dietary Histidine to Counteract Lymphatic and Mucosal Immune Dysfunction Due to Space Dysbiosis		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	wcromer@tamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	254-421-1530
Organization Name:	Texas A&M University		
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City:	Houston	**State:**	TX
Zip Code:	77030	**Congressional District:**	7
Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	03/01/2019	**End Date:**	06/30/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/30/2021 per NSSC information (Ed., 6/9/21) NOTE: End date changed to 2/28/2021 per NSSC information (Ed., 8/10/2020) NOTE: End date changed to 8/31/2020 per NSSC information (Ed., 8/3/2020)		
Key Personnel Changes/Previous PI:	September 2021 report: Dr. Robert Alaniz did not work on the project during the last year and so was no longer CoInvestigator.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0438		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The goal of our project was to determine what effect the space flight microbiome had on host immunity and physiological function. There were several objectives associated with this goal. The first was to determine if there were any major metabolomic changes in the gastrointestinal (GI) space flight microbiome and if there were significant shifts in the populations of the samples we were using. Our second objective was to determine if there were major genetic alterations in the immune/lymphatic systems from the same animals. The third objective was to determine if fecal microbiome transplant can recapitulate space flight changes we have observed in the immune and lymphatic systems. Our final objective was to determine if histidine supplementation could reverse any changes seen in fecal transplant. Significance: Determining the impact of the space flight microbiome on intestinal physiology and immunology is critical to understanding future risks that may impact human space exploration. Additionally development of a countermeasure to mitigate these issues is also necessary. This proposal addresses both of these critical issues utilizing a technique that has not yet been employed in the field of space biology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Determining the impact of the space flight microbiome/metabolome on the physiology and immunology of animals that have never been to space may seem to only have applications to space exploration but that is incorrect. The space flight environment is an extreme environment with a combination of microgravity and radiation and we (the scientific community) have very little information about the impact of these sorts of environments on both the microbiome and if those changes are clinically relevant. Therefore our studies have a crucial role in determining whether or not we should focus on therapeutic interventions focused on the microbiome of persons exposed to extreme environments (military, natural, and manmade disasters). This study will provide the basic foundation for making this determination.
Task Progress:	At the close of the project we have completed all the necessary animal work associated with this project. We have successfully completed the assessment of lymph transport and alterations in the immune system due to microbiome transplant and histidine supplementation. We are waiting on the delivery of RNASeq and metabolomic data from the Texas A&M cores (samples delivered >6 months ago). Once this data is in hand we will be able to make all final conclusions. From our initial findings the space flight microbiome most definitely conferred an immune and metabolic phenotype. There was minimal physiological alteration to the lymphatic system.
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Articles in Peer-reviewed Journals	Baranwal G, Creed HA, Cromer WE, Wang W, Upchurch BD, Smithhart MC, Vadlamani SS, Clark MC, Busbuso NC, Blais SN, Reyna AJ, Dongaonkar RM, Zawieja DC, Rutkowski JM. "Dichotomous effects on lymphatic transport with loss of caveolae in mice." Acta Physiol (Oxf). 2021 Aug;232(4):e13656. https://doi.org/10.1111/apha.13656 ; PMID: 33793057 , Aug-2021
Articles in Peer-reviewed Journals	Hongjiang S, Yin S, Wang W, Davies P, Sanchez E, Suntravat M, Zawieja D, Cromer WE. "Effect of Crotalus oreganus derived crotamine on lymphatic endothelial cells and lymph transport. " The FASEB Journal. 2020 Apr 18;34(S1). https://doi.org/10.1096/fasebj.2020.34.s1.06284 , Apr-2020